# Crypto Cash-and-Carry: Capital Efficiency in the Spot–Perpetual Basis Trade

## Summary

**Question.** Buying spot crypto and shorting the perpetual future ("perp") collects the perp's funding rate with no price exposure: economically, a secured loan of cash. Can the trade be improved by forecasting funding, or by optimising which coins to hold and how much?

**Approach.** Free exchange data for 274 coins on Binance and Bybit (2021–2026, including later-delisted coins), each exchange run as its own book. Every test had its decision rule written down before it ran, and every variant is counted when judging statistical significance.

**Findings.**

- **Carry over cash collapsed after 2021.** The exchange's funding floor (10.95% a year) stayed fixed while T-bills rose from 0% to 5%.
- **Funding is forecastable, but the edge is about the size of trading costs.**
- **No active strategy beat holding BTC and ETH passively**, including a daily MILP optimiser.
- **Capital structure decides the outcome.** The trade breaks even at about 1.4× capital per $1 of notional. Under portfolio margin (about 1.05×), passive BTC+ETH earned **+1.9% a year over T-bills (Sharpe 4.2)**; with spot and margin held separately (1.5×), +0.4%.

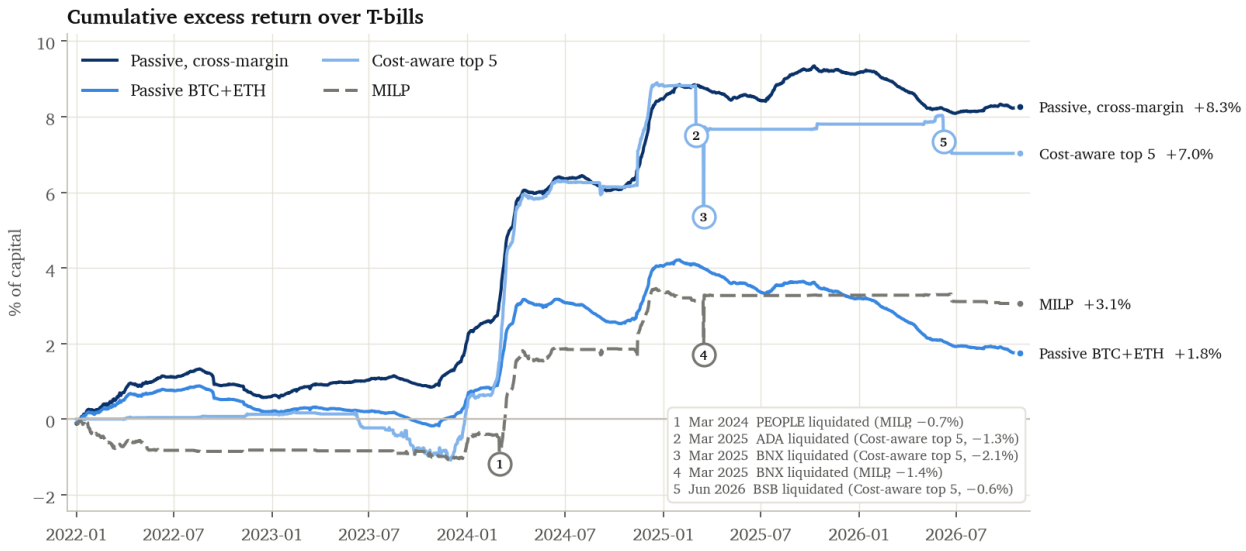



*Figure 1. Cumulative return over T-bills, $1M split across both exchanges, after all costs (2022–2026). Numbered dips are short-perp liquidations in price spikes on small coins (Section 4).*

## 1. Trade Mechanics

**Perpetual futures** never expire. Instead, longs and shorts exchange a **funding payment** every 1–8 hours that pushes the perp back towards the spot price:

- Perp above spot → longs pay shorts. Perp below spot → shorts pay longs.
- Binance's formula per 8 hours: **funding = premium + clamp(0.01% − premium, ±0.05%)**.
- For any small premium, the clamp returns exactly **0.01% per 8 hours = 10.95% a year**: an administered **floor**, much like an interest rate set by the exchange.

**The basis (cash-and-carry) trade:** buy $q$ coins of spot, short $q$ coins of the perp.

| P&L component | Source |
|---|---|
| Funding | received on the short perp each period (paid when funding is negative) |
| Basis change | gain or loss as the perp–spot gap moves |
| Trading costs | fees and slippage when opening, closing or resizing |
| Cost of capital | the T-bill interest given up on cash tied up in the position |

**Margin factor** = capital tied up per $1 of notional (spot plus margin for the short):

- **1.5** if spot and the short perp sit in separate accounts with a 50% buffer (the conservative case).
- **About 1.05** under portfolio margin, where spot counts as collateral at a 95% ratio.

**Excess return over T-bills ≈ funding ÷ margin factor − T-bill rate − trading costs.**

All results below are in excess of the 3-month T-bill, net of all costs.

## 2. Data

- **Exchanges:** Binance and Bybit (funding, daily volume, 8-hour spot and perp prices, 2021–2026); 3-month T-bill from FRED. Each exchange is a separate $500k book: every position is spot and perp on the same exchange.
- **Universe:** each month, the top 30 coins by trailing Binance volume among coins on both exchanges, using only data available at the time: **274 coins**, including coins later delisted.
- **Investable:** in that month's universe, with a spot market on the same exchange and a perp–spot gap within ±20%. About **25 coins per exchange** from 2022.
- **Periods:** 2021 trains the forecasts; all backtests run 2022–2026 out of sample.
- Stock, ETF and commodity perps are excluded (Appendix A).

## 3. Funding Forecasts

**Setup** (Appendix B):

- Target: each coin's average funding over the **next 7 days**.
- Four models: a constant (the 0.01% floor), last value, an exponentially weighted average (EWMA) and a pooled linear regression on recent funding averages, the perp–spot gap and time spent at the floor.
- Walk-forward on an expanding window: refitted monthly on all data up to then, tested out of sample from 2022.

**Results** (out of sample, 2022–2026):

| | Floor | Last value | EWMA | Regression |
|---|---|---|---|---|
| Forecast error, mean absolute (bp per 8h), range over years | 1.06–2.17 | 0.96–2.00 | 0.82–1.68 | **0.72–1.39** |
| Ranking of coins: rank IC, range over years | — | 0.37–0.60 | **0.49–0.70** | 0.47–0.68 |
| Top 20% by forecast vs average coin (% a year, before costs) | — | 4.6–10.5 | 6.5–15.2 | 6.3–16.8 |

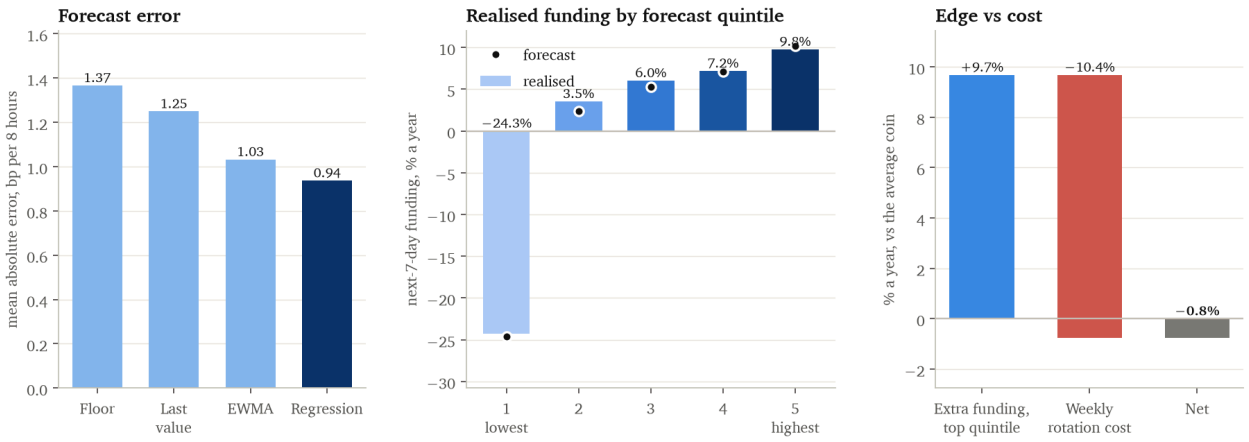



*Figure 2. Left: forecast error by model, 2022–2026. Middle: coins sorted each window into fifths by the regression forecast; bars show the funding they actually earned over the next 7 days, dots the forecast. Right: rotating weekly into the top fifth, against the average coin.*

- **The regression is the most accurate** (lowest error in 4 of 5 years) and is used downstream; the EWMA ranks coins slightly better.
- **Funding is clearly forecastable.** Realised funding rises steadily from the lowest forecast fifth (−24% a year) to the highest (+10%), and the forecasts sit close to the outcomes. A rank IC around 0.5 is very high (Appendix B).
- **The catch is cost.** The top fifth earns about 9.7% a year more than the average coin, but about half of it changes each week, and replacing it at 0.38% per round trip costs about 10.4% a year. Net of rotation, the edge is gone.

## 4. Strategies

All four run on each exchange separately, $500k each, at margin factor 1.5 unless stated (2022–2026).

| | Signal | Enter | Exit | Decides |
|---|---|---|---|---|
| **Passive** | none | hold BTC and ETH, always | only if the coin stops being investable | n/a |
| **Naive top 5** | 7-day forecast | the 5 coins with the highest forecast | drops out of the top 5 | every 7 days |
| **Cost-aware top 5** | 7-day forecast | forecast beats T-bills plus the round-trip cost spread over 7 days | forecast falls below T-bills | every 8 hours |
| **MILP** | 30-day forecast | chosen by the optimiser, net of trading costs | chosen by the optimiser | daily |

- **Passive** is the benchmark: the basis trade on the two largest coins, no selection.
- **Naive top 5** tests the forecast with no regard for cost.
- **Cost-aware top 5** enters only when the forecast clears the T-bill plus trading cost, and exits only when it falls below the T-bill. The gap between the two bars stops it trading in and out.
- **MILP** (mixed-integer program, solved with HiGHS) decides daily which coins to hold and how much of each. It maximises expected funding over 30 days, minus the T-bill on capital used, trading costs and a risk penalty for small and new coins, subject to the exchange's capital, per-coin caps (25% BTC/ETH, 10% other coins, 5% coins listed under 90 days), 1% of daily volume, a $10k minimum and 6 positions (Appendix D). The 30-day horizon spreads the round trip over a typical holding period, lowering the cost hurdle from about 20% to 4.6% a year.
- **Common rules:** forecasts are recomputed every 8 hours (the regression is refitted monthly); decisions are executed at the next 8-hour close; positions close on liquidation or when the coin leaves the universe.

**Results:**

| | Margin | Excess return a year | Sharpe | Max drawdown | Turnover (× capital a year) | Costs a year |
|---|---|---|---|---|---|---|
| **Passive BTC+ETH** | 1.50 | +0.36% | 1.10 | −2.5% | 0.1 | 0.03% |
| Naive top 5 | 1.50 | −4.31% | −1.68 | −19.0% | 31.3 | 5.95% |
| Cost-aware top 5 | 1.50 | +1.46% | 0.83 | −3.5% | 3.0 | 0.59% |
| Cost-aware top 5, EWMA forecast | 1.50 | +2.56% | 1.35 | −2.3% | 2.4 | 0.44% |
| MILP | 1.50 | +0.63% | 0.55 | −1.7% | 5.2 | 1.05% |
| MILP with turnover controls* | 1.05 / 1.30 | +1.45% | 1.21 | −2.6% | 4.2 | 0.84% |
| **Passive BTC+ETH, portfolio margin** | **1.05** | **+1.94%** | **4.18** | −1.2% | 0.2 | 0.04% |

** Exit costs charged in the objective, a no-trade band (trades smaller than $5k or 10% skipped), and realistic portfolio margin (1.05 for BTC/ETH, 1.30 for other coins).*

No active strategy beats passive at the same margin on a risk-adjusted basis (cost-aware rule P(not better) = 0.51; MILP 0.66; MILP with turnover controls 0.99), and none survives the correction for the number of variants tried (Appendix B).

**Why none of them beats passive:**

- **Naive rotation is eaten by costs:** the most funding collected (5.8% of capital a year) but 6.0% paid in costs and 0.6% in liquidation penalties.
- **The cost-aware rule mostly sits in cash:** a 0.38% round trip spread over 7 days requires forecast funding of about 24% a year to enter, so it held 1.7 coins on average out of 10 slots. Its higher return comes with more volatility, so its Sharpe is lower.
- **The MILP gives the carry back in trading.** It is fully invested in the high-funding periods of 2022 and 2024 and in T-bills otherwise (Figure 3), but makes about 90 opens and closes a year as forecasts change (Figure 4). Turnover controls cut that to about 65 and, with portfolio margin, more than doubled the return, still well short of passive under the same margin.
- **Small coins bring liquidations and basis losses.** Every numbered dip in Figure 1 is a short perp liquidated in a price spike on a small coin: PEOPLE (March 2024), ADA (March 2025, when it jumped on the US "crypto strategic reserve" announcement), BNX (March 2025, −2.1% in a day, mostly reversed the next day) and BSB (June 2026). Each costs a 2% penalty plus a loss on the leftover spot, sold at the next 8-hour close after the spike has partly reversed. BTC and ETH were never liquidated.
- **Capital cost is the largest drag in every strategy** (Figure 5). Passive BTC+ETH trades almost never and, under portfolio margin, ties up the least capital.

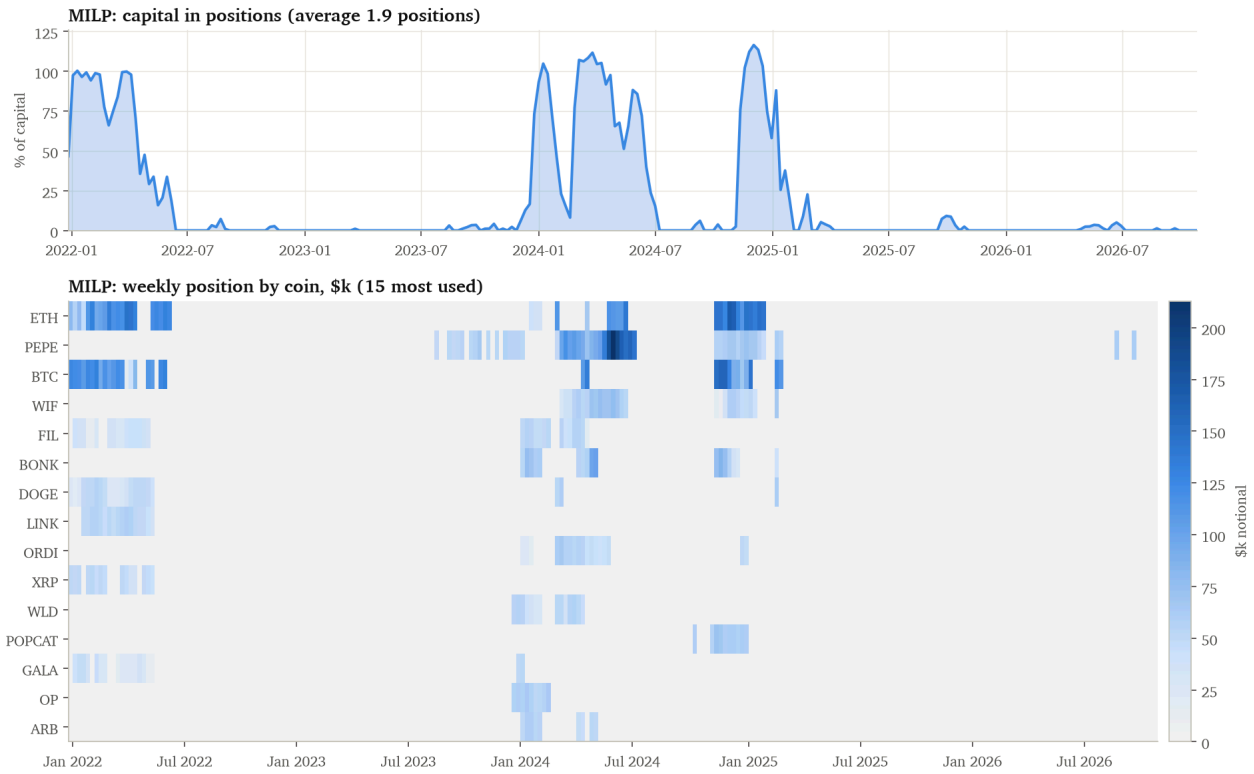



*Figure 3. MILP capital deployed and weekly positions by coin (both exchanges).*

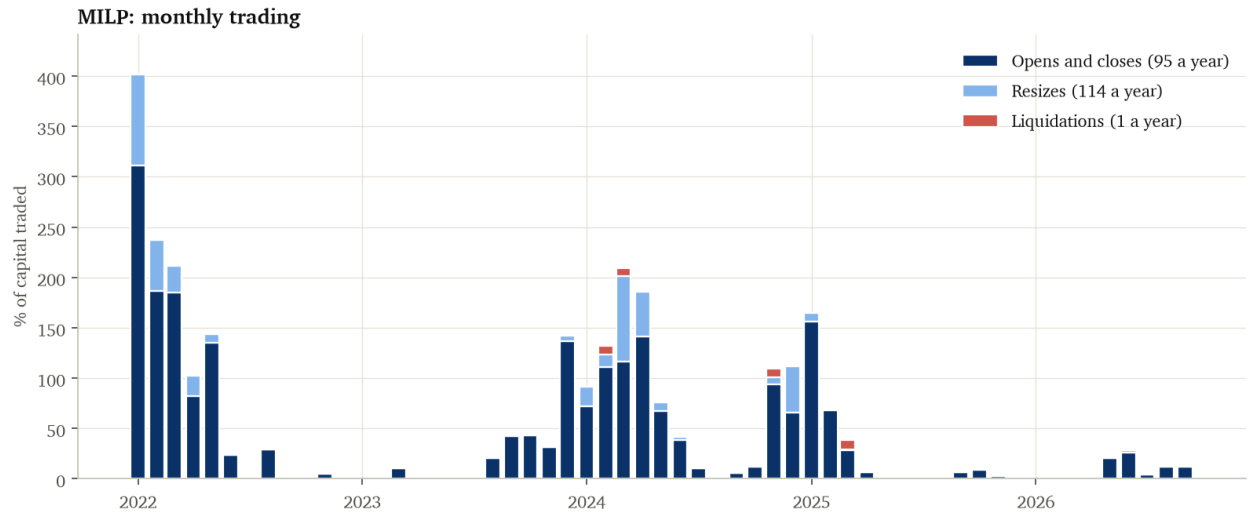



*Figure 4. MILP trading by month: opens and closes dominate, concentrated in the high-funding periods.*

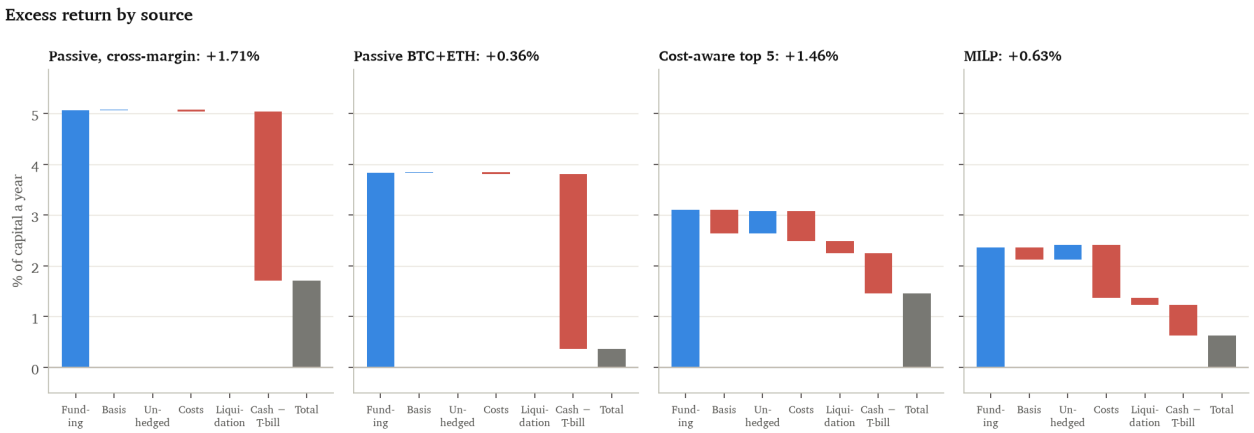



*Figure 5. Annual P&L by source. Blue adds, red subtracts, gray is the total. "Cash − T-bill" is the interest given up on capital tied up in positions.*

## 5. Carry Regimes

With selection ruled out, the return is the carry itself, and how much of it survived 2021 depends on the regime:

| Year | Context | BTC funding (% a year) | Median coin funding | T-bill | Funding prints negative |
|---|---|---|---|---|---|
| 2021 | Bull market, retail leverage | 30.6 | 10.95 | 0.05 | 8% |
| 2022 | LUNA, 3AC/Celsius, FTX collapses | 4.2 | 2.9 | 2.1 | 42% |
| 2023 | Recovery; policy rate about 5% | 7.9 | 10.95 | 5.3 | 19% |
| 2024 | Spot BTC ETFs, rally | 11.9 | 10.95 | 5.2 | 13% |
| 2025 | Weaker leverage demand | 5.1 | 6.75 | 4.2 | 29% |
| 2026 (to Oct) | BTC mostly below the floor | 3.1 | 4.8 | 3.8 | 33% |

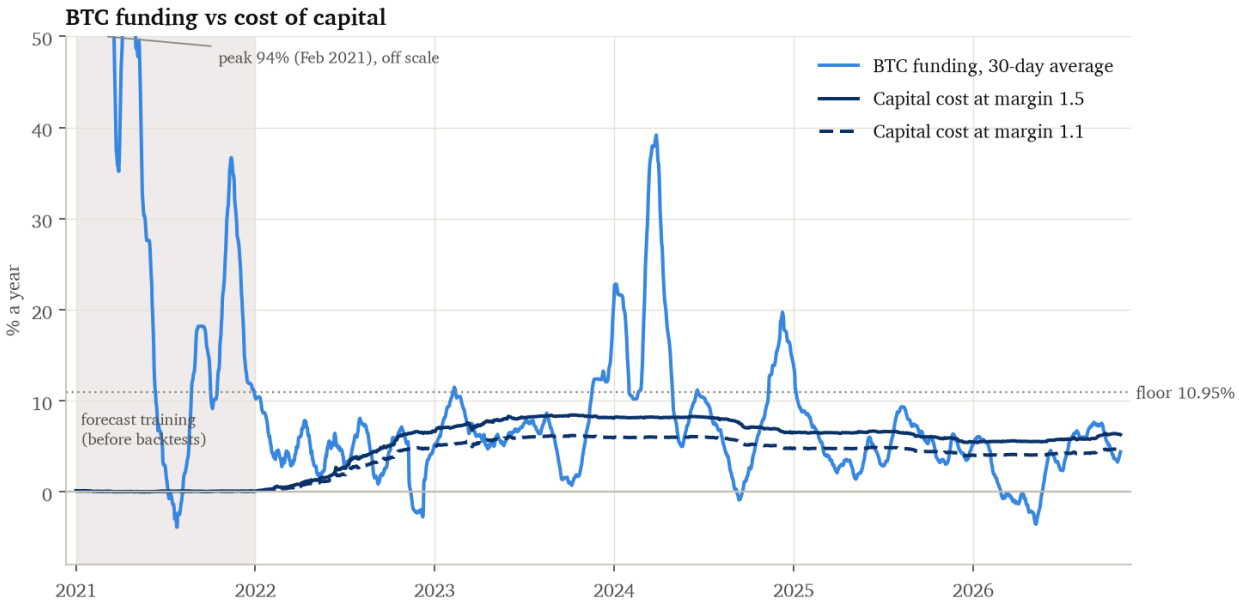



*Figure 6. BTC funding (30-day average) against the cost of capital, from the start of the data. The trade earns only when funding is above the capital-cost line: continuously in 2021, in bursts since.*

**Why carry compressed:**

- **The floor is set by the exchange, not by interest rates.** 10.95% a year was 11% over cash in 2021 and about 6% over cash once T-bills reached 5%; about 2.5% after the capital cost at margin factor 1.5.
- **Leverage demand is cyclical:** 30.6% BTC funding in 2021, negative 42% of the time in 2022.
- **The perp has traded below spot since 2022** (median −2 to −6 bp): more natural short sellers than leveraged buyers.
- **More arbitrage capital** (spot ETFs and Ethena's USDe stablecoin in 2024) is a likely contributor, not tested here.
- **Fewer coins can be hedged:** new coins increasingly list as perps only, so spot is available for only 72–78% of the top 30 by 2026, down from 100%.

In 2021 alone, before the backtest period, passive BTC+ETH earned +12.6% over T-bills; it is not counted in any result above, since the forecasts needed that year to train.

## 6. Margin and Cost Sensitivity

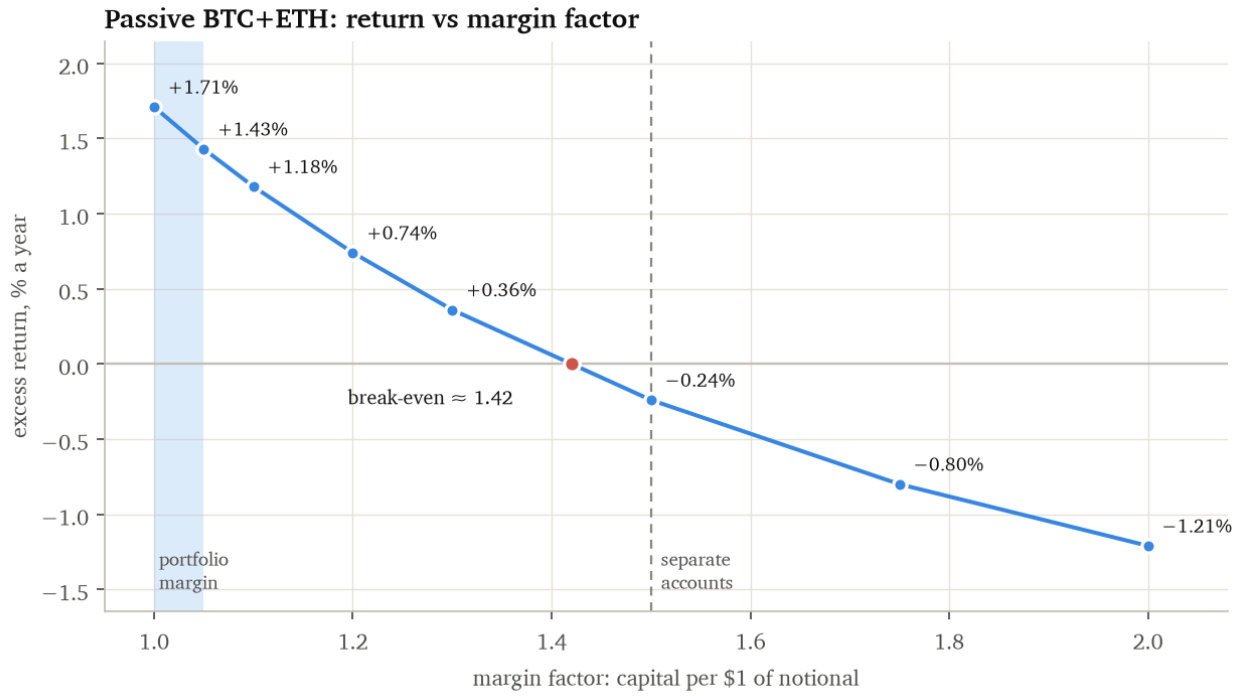



*Figure 7. Passive BTC+ETH excess return against capital tied up per $1 of notional (constant exposure, weekly rebalance). Break-even is about 1.4.*

**Margin is the main lever.**

- Each additional 0.1 of margin factor costs about 0.4–0.5% a year.
- Break-even ≈ funding ÷ (T-bill + costs) ≈ 6.4% ÷ 4.6% ≈ **1.4**.
- Exchange documentation puts realistic BTC/ETH portfolio margin at 1.0–1.05 (95% collateral ratio, 0.4% maintenance margin).
- Earning interest on the capital already tied up (yield-bearing collateral) would attack the largest cost directly.

**Costs matter most for active strategies.** All fees and slippage multiplied by k (k = 1 is the base case, about 0.38% per round trip):

| k | 0.25 | 0.5 | 1 | 2 |
|---|---|---|---|---|
| Passive BTC+ETH (margin 1.10) | +1.56% | +1.43% | +1.18% | +0.68% |
| MILP (margin 1.50) | +1.12% | +0.82% | +0.63% | +0.28% |

Even with maker or VIP fees, the MILP stays below passive under the same margin.

## 7. Conclusion

- The carry that survived 2021 is a **financing spread**: funding near an exchange-set floor, earned against the cost of cash.
- It cannot be improved by coin selection (the edge is about the cost of trading it) or by a constrained optimiser (it gives the carry back in turnover).
- What moves returns is the balance sheet: **the margin factor, whether collateral earns interest, and execution cost**.

**Limitations**

- Five out-of-sample years, with most of the P&L in 2024. Low-volatility carry flatters the Sharpe ratio: exchange failure, stablecoin de-pegs and liquidation cascades barely appear in five years of daily data.
- 8-hour prices only; flat base-tier fees; an approximate liquidation rule; margin modelled as a single factor.
- No historical coin-borrow data, so the reverse trade (long perp, short spot) is untested.
- Neither Binance nor Bybit serves Ontario residents: this describes the market, not an accessible strategy.

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## Appendix A. Data and Universe

- **Sources:** Binance USD-M API and bulk archive (data.binance.vision), used for delisted symbols the live API no longer serves; Bybit v5 API (spot only from July 2021); FRED for the T-bill, lagged one day.
- **Universe:** monthly top 30 by trailing 30-day Binance perp volume among coins traded on both exchanges on at least 25 of the previous 30 days. Stablecoins excluded. Stock, ETF and commodity perps (Binance tag `TRADIFI_PERPETUAL`) excluded; by 2026 they held about a third of top-30 volume.
- **Common grid:** exchanges settle some coins every 1 or 4 hours, so all funding is summed into 8-hour windows ending 00:00, 08:00 and 16:00 UTC. Prices are 8-hour closes; perp prices are divided by contract multipliers (e.g. 1000PEPE).
- **Investable:** spot market on the same exchange, and a perp–spot gap within ±20% (larger gaps are collapses or broken pairs, e.g. LUNA in May 2022). Under 0.5% of windows are missing.

## Appendix B. Research Protocol

**Pre-registration.** Each test's design and pass/fail rule was written down before it ran:

| Test | Rule | Outcome |
|---|---|---|
| Forecast model choice | Regression used only if it beats last value and EWMA on error in ≥ 4 of 5 years | Pass (4 of 5) |
| Forecast-driven selection | Cost-aware top 5 beats passive Sharpe ($P < 0.10$) and is positive in ≥ 3 of 5 years | Fail ($P = 0.51$) |
| MILP | Beats passive Sharpe ($P < 0.10$) and positive in ≥ 3 of 5 years | Fail ($P = 0.66$) |
| MILP with turnover controls | Beats passive at the same margin ($P < 0.10$), positive in ≥ 3 of 5 years | Fail ($P = 0.99$) |

The MILP originally allocated across both exchanges jointly and could move capital between them. Fixing the split at 50/50 changed its return only from +0.67% to +0.60% a year, so it was simplified to one book per exchange after these results; all earlier runs still count as trials.

**Statistics used:**

- **Paired block bootstrap:** resamples blocks of days from two strategies together, to test whether one Sharpe ratio is really higher than another.
- **Deflated Sharpe ratio:** the probability that a strategy's Sharpe is real after accounting for how many variants were tried (23 here). The best of many skill-free strategies looks good by luck; this measure asks whether a result beats that luck benchmark. Only passive BTC+ETH under portfolio margin survives.
- **Rank IC:** at each point in time, the correlation between how coins are ranked by forecast and how they rank by actual funding afterwards, averaged over time. Zero is no information; above 0.05 is useful in equity research; the forecasts here reach about 0.5 because funding is persistent.

**Forecast details:**

- Features: last funding; EWMAs with 1, 7 and 30-day half-lives; perp–spot gap; share of the last 3 days at the floor.
- The EWMA model's half-life is chosen from {1, 3, 7, 30} days at each monthly refit, using training data only.
- No look-ahead: a training row is used only if its whole target window ends before the refit date. A test scrambles all data after a cut-off and checks that no earlier forecast changes.

## Appendix C. Backtest Engine

- **Timing:** decide at an 8-hour close using data up to then; trade at the next close.
- **P&L per window:** funding + basis change + cash interest − costs − liquidation penalties. The components add up to the equity change exactly (tested).
- **Costs (base tier, taker):** spot 0.10%; perp 0.05% (Binance), 0.055% (Bybit); plus 2 bp slippage per leg: about 0.38% per round trip.
- **Margin:** 50% buffer on the short, no cross-margining (margin factor 1.5) unless stated; margin topped up once a day.
- **Liquidation:** if the perp rises more than 40% since the last daily top-up, the short is closed with a 2% penalty and the spot is sold at the next close.
- **Exits:** coins that stop being investable are closed with full costs; coins with no price for 3 days (delisted) are closed at their last prices.
- **Liquidity:** positions capped at 1% of 7-day average daily volume.

## Appendix D. MILP Formulation

Solved separately for each exchange, with capital C. For each candidate coin, with current notional $x^0$ and forecast annual funding f:

- **Variables:** target notional $x \geq 0$; buys $b \geq 0$; sells $s \geq 0$ ($s \leq x^0$); open position $y \in \{0, 1\}$.
- **Objective (maximise, over a horizon H of 30 days):** $\Sigma\, x \cdot (f - m \cdot r - \text{penalty}) \cdot H - \Sigma\, b \cdot$ round-trip cost, where m is the margin factor, r the T-bill rate, and the penalty is 0% (BTC/ETH), 3% (other coins) or 10% (coins listed under 90 days) a year.
- **Constraints:**
- $x = x^0 + b - s$;
- $x \leq \text{cap} \cdot y$ and $x \geq \$10\text{k} \cdot y$, where cap = min(per-coin share of C, 1% of daily volume);
- $\Sigma\, y \leq 6$;
- $\Sigma\, m \cdot x \leq C$.
- **Turnover-controlled version:** sells also pay the exit half of the round trip, and each held position either stays unchanged or trades by at least max($5k, 10% of the position). A buy-or-sell binary per coin stops a same-day buy and sell from meeting that minimum while barely moving the position.
- Solved daily with HiGHS (via `scipy.optimize.milp`); positions left unchanged are held as is rather than re-traded as prices drift.

## Appendix E. Corrections

Errors found after first results were fixed, tested and rerun; no decision changed.

| Found by | Problem | Effect of the fix |
|---|---|---|
| Engine refactor | A coin with spot but no perp price produced undefined equity, silently dropping months of one variant | EWMA variant Sharpe 2.34 → 1.35 |
| 30-day forecasts | Look-ahead gap hard-coded to 7 days | Prevented a leak before any 30-day result |
| Position charts | Delisted coins (e.g. MATIC after its rename) held forever with no price | Active strategies improved |
| Trade counts | Optimiser positions re-traded as prices drifted | MILP resizes fell from ~615 to ~105 a year |
| Data end date | Final window lacked a funding print, forcing an artificial exit | Passive Sharpe 1.01 → 1.10 |
| Unit test | The no-trade band could be met by buying and selling the same coin on one day | Fixed with a buy-or-sell binary |

## Appendix F. Reproduction

```
python -m venv .venv && .venv/bin/pip install -e ".[dev]"
.venv/bin/python research/build_data.py              # data (~1.5 h the first time;
resumable)
.venv/bin/python research/phase1_descriptives.py     # descriptives (funding regimes,
floor, coverage)
.venv/bin/python research/phase2_forecast.py         # investable set, forecasts
.venv/bin/python research/phase3_backtest.py         # coin-selection rules
.venv/bin/python research/phase4_allocator.py        # MILP
.venv/bin/python research/phase4b_allocator.py       # MILP with turnover controls
.venv/bin/python research/sensitivity.py             # margin and costs (also leverage,
not in the paper)
.venv/bin/python research/figures.py                 # figures
.venv/bin/pytest                                     # tests
```

Detailed write-ups per stage: `docs/phase1.md` to `docs/phase4b.md`, `docs/sensitivity.md`. Decision rules: `docs/plan.md`. Every decision and correction: `docs/decisions.md`.